Submission C solution to Problem 5

Abstract

We prove that the Markov semigroup associated with the Dirichlet stochastic heat equation with drift $\delta_0(u)$ has at most one invariant probability measure. In fact, under the hypotheses stated in the question, the invariant measure is unique and is the Gibbs perturbation of the Brownian bridge measure

$$\mu(\mathrm{d}u) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{u(\xi)>0\}} \, \mathrm{d}\xi\right) \gamma(\mathrm{d}u),$$

where γ is the law of the standard Brownian bridge on $[0,1]$. The proof uses three inputs: the reversible Gibbs structure of smooth approximations, the absolute-continuity theorem of Bounebache–Zambotti for the associated Dirichlet-form resolvent, and irreducibility of the weighted Ornstein–Uhlenbeck Dirichlet form.

1 Notation and the candidate invariant measure

Let

$$E := C_0([0,1]), \qquad H := L^2(0,1).$$

We regard E as a Borel subset of H. Let γ denote the law on E of the standard Brownian bridge from 0 to 0 over $[0,1]$. Equivalently, γ is the centered Gaussian measure on H with covariance $(-\Delta_D)^{-1}$, where Δ_D is the Dirichlet Laplacian on $(0,1)$.

Let

$$\mathsf{H}(r) := \mathbf{1}_{\{r>0\}}, \qquad \Psi(u) := \int_0^1 \mathsf{H}(u(\xi)) \, \mathrm{d}\xi, \qquad u \in E.$$

Define

$$\mu(\mathrm{d}u) := Z^{-1} e^{2\Psi(u)} \gamma(\mathrm{d}u), \qquad Z := \int_E e^{2\Psi(u)} \gamma(\mathrm{d}u). \tag{1.1}$$

Since $0 \le \Psi \le 1$, the measures μ and γ are equivalent and their Radon–Nikodym derivative is bounded above and below by positive constants.

For $n \ge 1$, let

$$G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} \, e^{-nr^2/2}, \qquad B_n(r) := \int_{-\infty}^r G_{1/n}(a) \, \mathrm{d}a,$$

and put

$$\Psi_n(u) := \int_0^1 B_n(u(\xi)) \, \mathrm{d}\xi, \qquad \mu_n(\mathrm{d}u) := Z_n^{-1} e^{2\Psi_n(u)} \gamma(\mathrm{d}u).$$

Then $B_n' = G_{1/n}$.

Lemma 1.1. *The measures μ_n converge to μ in total variation.*

Proof. For a Brownian bridge β,

$$\mathbb{E}_\gamma\left[\int_0^1 \mathbf{1}_{\{\beta(\xi)=0\}} \, \mathrm{d}\xi\right] = \int_0^1 \gamma(\beta(\xi) = 0) \, \mathrm{d}\xi = 0,$$

because $\beta(\xi)$ is a non-degenerate centered Gaussian random variable for $\xi \in (0,1)$, while the endpoints have Lebesgue measure zero. Hence, for γ-a.e. u, the zero set $\{\xi : u(\xi) = 0\}$ has Lebesgue measure zero. Since $B_n(r) \to \mathsf{H}(r)$ for every $r \neq 0$ and $0 \leq B_n \leq 1$, dominated convergence gives $\Psi_n(u) \to \Psi(u)$ for γ-a.e. u. Another application of dominated convergence gives $Z_n \to Z$ and

$$Z_n^{-1} e^{2\Psi_n} \longrightarrow Z^{-1} e^{2\Psi} \quad \text{in } L^1(\gamma).$$

This is exactly total-variation convergence. □

2 Reversibility of the limiting semigroup

Let P_t^n be the Markov semigroup for the regularized equation

$$\partial_t u^n = \frac{1}{2} \partial_{xx} u^n + G_{1/n}(u^n) + \dot{W}$$

with Dirichlet boundary conditions. The following is standard for gradient-type stochastic heat equations with smooth drift.

Lemma 2.1. *For each n, μ_n is invariant and reversible for P_t^n:*

$$\int_E f\, P_t^n g \; \mathrm{d}\mu_n = \int_E g\, P_t^n f \; \mathrm{d}\mu_n$$

for all bounded Borel $f, g : E \to \mathbb{R}$ and all $t \geq 0$.

Proof. For cylinder functions φ, ψ on H, the generator of the regularized equation is

$$L_n \varphi = L_0 \varphi + \langle D\Psi_n, D\varphi \rangle_H,$$

where L_0 is the Ornstein–Uhlenbeck generator associated with the linear Dirichlet stochastic heat equation. Since $\mu_n = Z_n^{-1} e^{2\Psi_n} \gamma$, the Gaussian integration-by-parts formula gives

$$\int \psi\, L_n \varphi \; \mathrm{d}\mu_n = -\frac{1}{2} \int \langle D\varphi, D\psi \rangle_H \; \mathrm{d}\mu_n = \int \varphi\, L_n \psi \; \mathrm{d}\mu_n.$$

Thus the closure of L_n is self-adjoint in $L^2(\mu_n)$, and the associated Markov semigroup is symmetric. Taking $\psi \equiv 1$ gives invariance. This is the usual Gibbs-reversibility computation for stochastic quantization; see, for example, Bounebache–Zambotti [BZ14, Section 4] for the same calculation in the present Dirichlet setting. □

We now pass to the singular limit.

Lemma 2.2. *The measure μ is invariant and reversible for P_t. Namely, for all bounded Borel $f, g : E \to \mathbb{R}$ and all $t \geq 0$,*

$$\int_E f\, P_t g \; \mathrm{d}\mu = \int_E g\, P_t f \; \mathrm{d}\mu. \tag{2.1}$$

Proof. First take $f, g \in C_b(E)$. By the assumed weak convergence of the regularized solutions,

$$P_t^n h(u) \longrightarrow P_t h(u), \qquad u \in E,$$

for every $h \in C_b(E)$ and every fixed $t > 0$. Therefore, using Lemma 1.1,

$$\left| \int f P_t^n g \; \mathrm{d}\mu_n - \int f P_t g \; \mathrm{d}\mu \right| \leq \|f\|_\infty \|g\|_\infty \, \|\mu_n - \mu\|_{\mathrm{TV}} + \int |f(u)|\, |P_t^n g(u) - P_t g(u)|\, \mu(\,\mathrm{d}u) \longrightarrow 0.$$

Similarly,

$$\int g P_t^n f \, \mathrm{d}\mu_n \longrightarrow \int g P_t f \, \mathrm{d}\mu.$$

The reversibility identity for P_t^n therefore passes to the limit and yields (2.1) for bounded continuous f, g. Since E is Polish and μ is a probability measure, a monotone-class argument extends (2.1) to all bounded Borel f, g. Taking $g \equiv 1$ gives invariance of μ. $\square$

3 The Dirichlet form and its irreducibility

Let $\mathcal{F}\mathcal{C}_b^1$ denote the smooth bounded cylinder functions on H. Define on $\mathcal{F}\mathcal{C}_b^1$ the bilinear form

$$\mathcal{E}(F, G) := \frac{1}{2} \int_E \langle DF(u), DG(u) \rangle_H \, \mu(\mathrm{d}u). \tag{3.1}$$

Because $\mathrm{d}\mu / \mathrm{d}\gamma$ is bounded above and below by positive constants, (3.1) is closable in $L^2(\mu)$. We denote its closure by $(\mathcal{E}, \mathcal{F})$.

Lemma 3.1. *The symmetric semigroup $(P_t)_{t \geq 0}$ is the Markov semigroup associated with $(\mathcal{E}, \mathcal{F})$. Moreover, $(\mathcal{E}, \mathcal{F})$ satisfies a Poincare inequality: there exists $C < \infty$ such that for every $F \in \mathcal{F}$,*

$$\mathrm{Var}_\mu(F) \leq C \, \mathcal{E}(F, F). \tag{3.2}$$

Consequently, if $F \in L^2(\mu)$ and $P_t F = F$ in $L^2(\mu)$ for all $t \geq 0$, then F is μ-a.s. constant.

Proof. For $f = -2\mathsf{H}$, Bounebache–Zambotti [BZ14, Propositions 2.4 and 3.1] construct the quasi-regular Dirichlet form

$$\frac{1}{2} \int \langle DF, DG \rangle \, \mathrm{d}\nu_f, \qquad \nu_f(\mathrm{d}u) = \frac{1}{Z_f} \exp\Big(- \int_0^1 f(u(\xi)) \, \mathrm{d}\xi \Big) \gamma(\mathrm{d}u).$$

Since $f = -2\mathsf{H}$, we have $\nu_f = \mu$. For smooth f, their local-time formulation is equivalent, by the occupation-time formula, to

$$\partial_t u = \frac{1}{2} \partial_{xx} u - \frac{1}{2} f'(u) + \dot{W}.$$

For $f = -2\mathsf{H}$, one has $-\frac{1}{2} f' = \delta_0$ in the sense of distributions. Thus the Bounebache–Zambotti process solves the same singular SPDE. By the assumed uniqueness in law of the ABLM solution, the associated semigroup agrees with P_t. Hence P_t is the semigroup of $(\mathcal{E}, \mathcal{F})$.

It remains to prove (3.2). Let $\mathcal{E}_0$ be the Ornstein–Uhlenbeck Dirichlet form on $L^2(\gamma)$:

$$\mathcal{E}_0(F, F) = \frac{1}{2} \int \|DF\|_H^2 \, \mathrm{d}\gamma.$$

The Brownian-bridge Ornstein–Uhlenbeck form has a spectral gap; equivalently, there exists $C_0 < \infty$ such that

$$\mathrm{Var}_\gamma(F) \leq C_0 \, \mathcal{E}_0(F, F), \qquad F \in \mathrm{Dom}(\mathcal{E}_0).$$

This follows from the Hermite expansion of the Dirichlet Ornstein–Uhlenbeck generator; see Bounebache–Zambotti [BZ14, Proposition 2.1].

Since $\mathrm{d}\mu = m \, \mathrm{d}\gamma$ with $0 < a \leq m \leq b < \infty$, the form domains of $\mathcal{E}$ and $\mathcal{E}_0$ coincide and

$$a \, \mathcal{E}_0(F, F) \leq \mathcal{E}(F, F) \leq b \, \mathcal{E}_0(F, F).$$

Therefore,

$$\begin{aligned}\mathrm{Var}_\mu(F) &= \inf_{c\in\mathbb{R}} \int (F-c)^2 \,\mathrm{d}\mu \\ &\le \int \left(F - \int F \,\mathrm{d}\gamma\right)^2 \,\mathrm{d}\mu \\ &\le b\ \mathrm{Var}_\gamma(F) \le bC_0\,\mathcal{E}_0(F,F) \le \frac{bC_0}{a}\,\mathcal{E}(F,F).\end{aligned}$$

This proves (3.2). If $P_tF = F$ for all t, then

$$\mathcal{E}(F,F) = \lim_{t\downarrow 0} \frac{1}{t}\langle F - P_tF, F\rangle_{L^2(\mu)} = 0,$$

and (3.2) implies $\mathrm{Var}_\mu(F) = 0$. □

We also need the absolute-continuity property of the resolvent. For $\alpha > 0$, write the normalized resolvent as

$$\mathcal{R}_\alpha(x,A) := \alpha \int_0^\infty e^{-\alpha t} P_t(x,A)\,\mathrm{d}t, \qquad x \in E,\ A \in \mathcal{B}(E).$$

Lemma 3.2 (Resolvent absolute continuity)**.** *For every $\alpha > 0$ and every $x \in E$,*

$$\mathcal{R}_\alpha(x,\cdot) \ll \mu.$$

Proof. Bounebache–Zambotti prove the absolute-continuity condition for the resolvent of the Dirichlet-form process associated with ν_f for every bounded function f of bounded variation; see [BZ14, Proposition 2.5]. Applying their result to $f = -2\mathsf{H}$ gives absolute continuity with respect to μ. Their Proposition 3.1 identifies the associated process as a weak solution of the corresponding local-time SPDE. As explained in the proof of Lemma 3.1, for $f = -2\mathsf{H}$ this is precisely the SPDE with drift $\delta_0(u)$. The assumed uniqueness in law of the ABLM mild solution therefore identifies this resolvent with the resolvent of P_t. Hence $\mathcal{R}_\alpha(x,\cdot) \ll \mu$. □

4 Uniqueness of the invariant measure

Theorem 4.1. *The semigroup $(P_t)_{t\ge 0}$ has exactly one invariant probability measure on $E = C_0([0,1])$. It is the measure μ defined in (1.1). In particular, P_t has at most one invariant probability measure.*

Proof. By Lemma 2.2, μ is invariant. Let λ be any invariant probability measure for P_t. We first show that $\lambda \ll \mu$.

Let $A \in \mathcal{B}(E)$ satisfy $\mu(A) = 0$. By Lemma 3.2, $\mathcal{R}_\alpha(x,A) = 0$ for every $x \in E$. Since λ is invariant,

$$\begin{aligned}\lambda(A) &= \alpha \int_0^\infty e^{-\alpha t} \lambda P_t(A)\,\mathrm{d}t \\ &= \int_E \mathcal{R}_\alpha(x,A)\,\lambda(\mathrm{d}x) = 0.\end{aligned}$$

Thus $\lambda \ll \mu$. Write $h := \mathrm{d}\lambda/\mathrm{d}\mu$. Then $h \ge 0$ and $\int h\,\mathrm{d}\mu = 1$.

We claim that $P_th = h$ in $L^1(\mu)$ for every $t \ge 0$. Indeed, for every bounded Borel φ,

$$\begin{aligned}\int_E \varphi\, P_th\,\mathrm{d}\mu &= \int_E P_t\varphi\, h\,\mathrm{d}\mu \\ &= \int_E P_t\varphi\,\mathrm{d}\lambda \\ &= \int_E \varphi\,\mathrm{d}\lambda \\ &= \int_E \varphi\, h\,\mathrm{d}\mu.\end{aligned}$$

Here the first equality uses the symmetry of P_t in $L^2(\mu)$, extended by duality to $L^1(\mu)$–$L^\infty(\mu)$, and the third equality uses invariance of λ.

Fix $N > 0$ and set $h_N := h \wedge N$. Since $r \mapsto r \wedge N$ is concave and nondecreasing, Jensen's inequality for the Markov operator P_t gives

$$P_t h_N \leq (P_t h) \wedge N = h_N.$$

Integrating with respect to the invariant measure μ,

$$\int P_t h_N \, \mathrm{d}\mu = \int h_N \, \mathrm{d}\mu.$$

Therefore $P_t h_N = h_N$ μ-a.s. for every $t \geq 0$. Since $h_N \in L^2(\mu)$, Lemma 3.1 implies that h_N is μ-a.s. constant.

Taking $N = 2$, we obtain $h \wedge 2 = c$ for some constant c. Since

$$c = \int (h \wedge 2) \, \mathrm{d}\mu \leq \int h \, \mathrm{d}\mu = 1 < 2,$$

the identity $h \wedge 2 = c$ forces $h = c$ μ-a.s. Finally, $\int h \, \mathrm{d}\mu = 1$ gives $c = 1$. Hence $\lambda = \mu$.

Thus μ is the unique invariant probability measure. □

Remark 4.2. *The key point excluding additional singular invariant measures is the pointwise absolute continuity of the normalized resolvent in Lemma 3.2. Symmetry and the Poincare inequality alone imply uniqueness only among invariant probability measures that are already absolutely continuous with respect to μ.*

References

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